

Post-Incident Report: Network Convergence Failure & Host Resource Exhaustion

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Subject: Layer 2 Loop Remediation and Environment Hardening

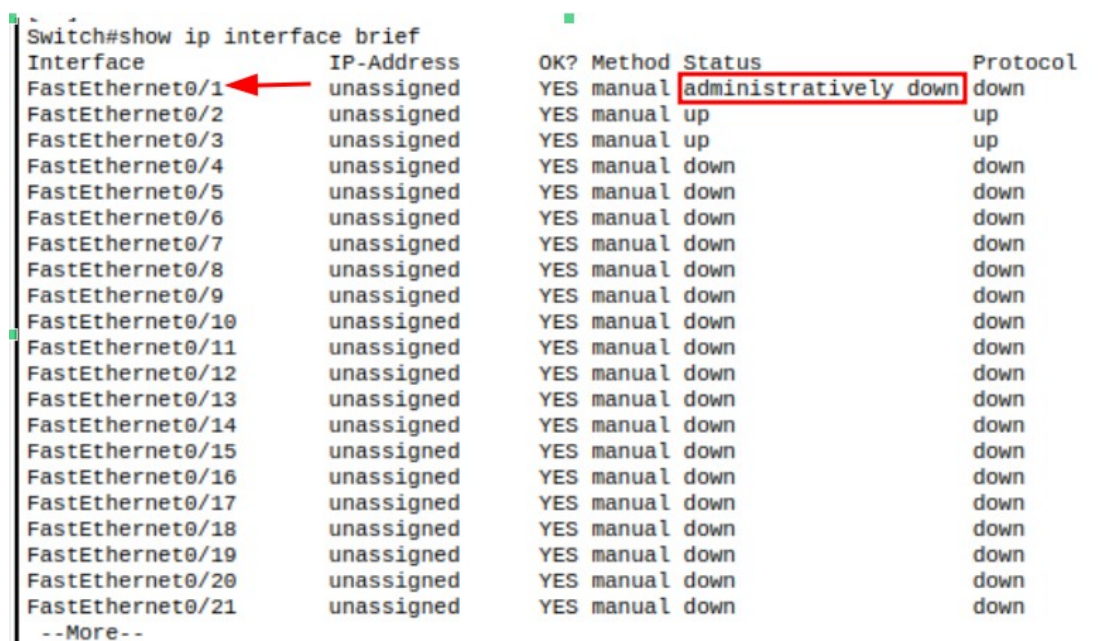
1. Incident Overview

At approximately 17:30 IST, the local engineering workstation experienced a **Kernel Panic/Total System Freeze**. This was identified as a physical-to-logical failure cascading from the simulation environment.

2. Root Cause Analysis (RCA)

The failure originated from an unintentional **Layer 2 Loop** between Switch-0 and Switch-1.

- **Mechanism:** In the absence of an active Spanning Tree Protocol (STP) convergence, a **Broadcast Storm** was generated.
- **Impact:** The infinite frame replication consumed the host machine's CPU cycles, leading to a total "Physical-to-Logical" crash and loss of unsaved volatile data.



```
Switch#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	administratively down	down
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down

--More--

Fig.1 A brief catalog of IP Interfaces using "show ip interface brief" command in Switch0's Command Line Interface / CLI.

3. Recovery & Remediation (I&R)

Following a hard reset of the host hardware, the following **Standard Operating Procedures (SOP)** were executed:

- **Administrative State Recovery:** Identified that the management interface was Administratively Down. Restored state via CLI.
- **Static Management Assignment:** Established a fixed management IP (192.168.1.1/24) to ensure **Incorruptible Governance** of the switch fabric.
- **STP Verification:** Confirmed protocol convergence. The system successfully elected a Root Bridge and designated a **Blocking Port** (Yellow LED), neutralizing the loop.

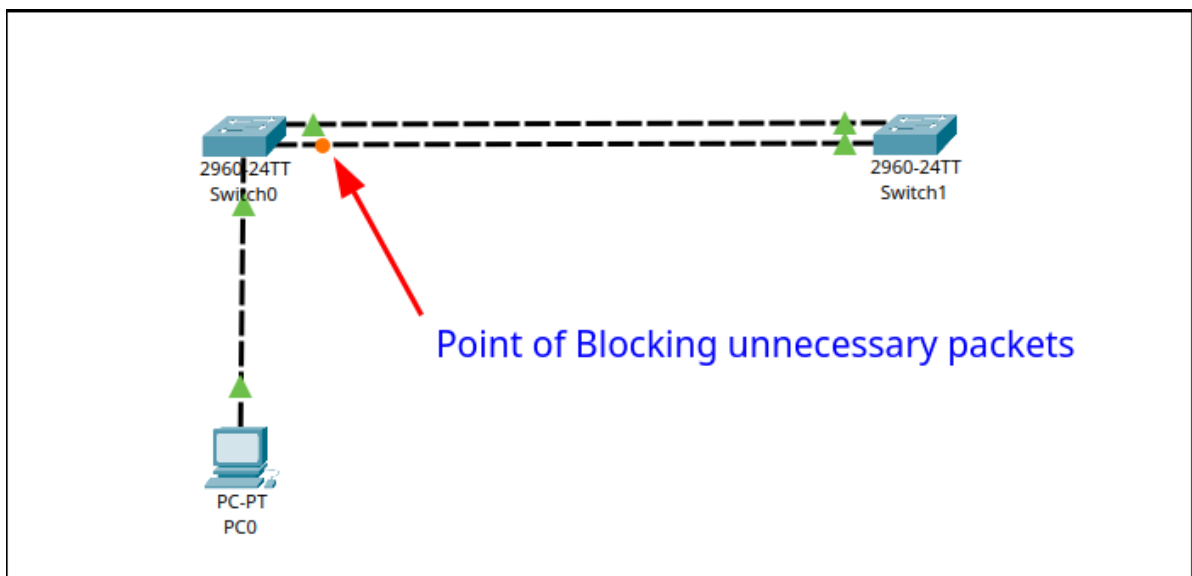
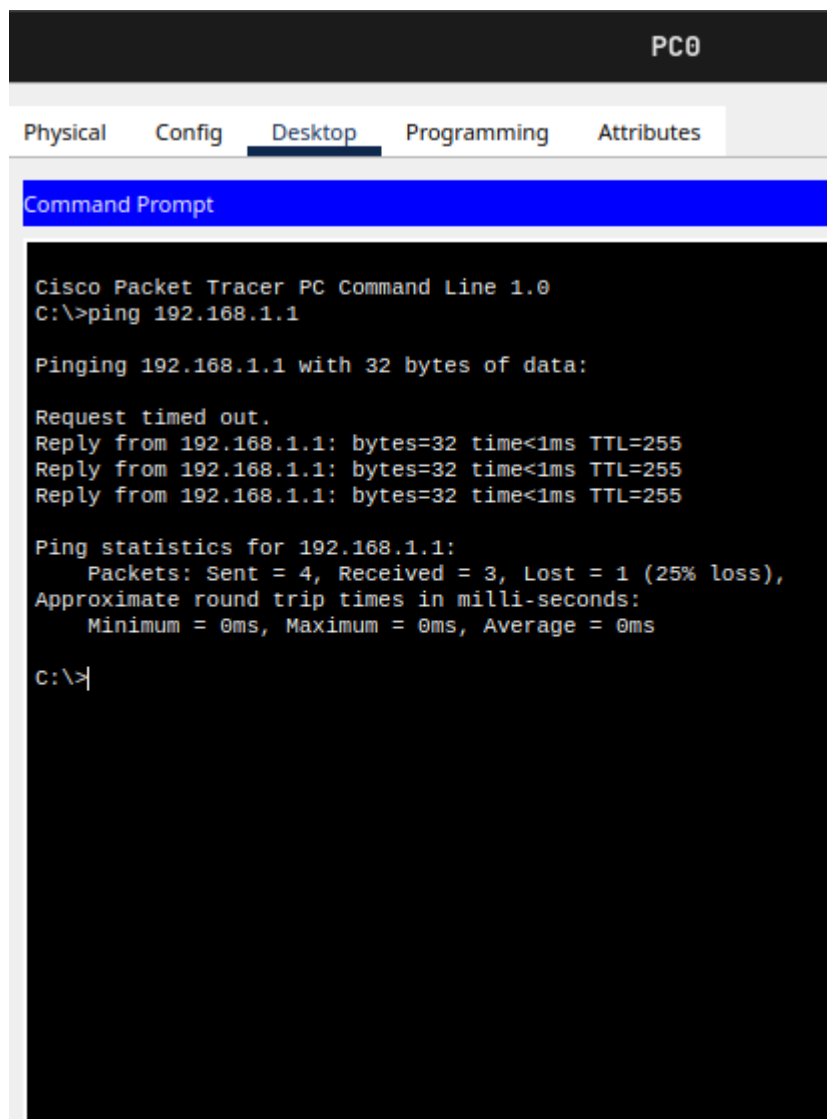


Fig.2 Network design with assembling of two Switches (Switch0 and Switch1) with one Personal Computer (PC0).



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the 'ping 192.168.1.1' command. The output indicates that the ping was successful, with 32 bytes of data being sent and received. The statistics show 4 packets sent, 3 received, and 1 lost (25% loss). The round trip times are all 0ms.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

c:\>|
```

Fig.3 Result after executing “ping 192.168.1.1” command in PC0’s Command Prompt.

4. Hardening & Prevention

To mitigate future resource exhaustion and maintain **Professional Integrity**:

- **Edge Security:** Enabled spanning-tree bpduguard enable on all access-layer ports to prevent unauthorized loops.
- **Persistence:** Committed all changes to NVRAM using the write memory command to ensure recovery survivability.
- **Documentation:** Technical artifacts (.pkt file and CLI logs) have been staged for the GitHub repository.